

Top 5 Related Research Papers

The works most central to slidekit's goals and code — with seven main points on each paper

How these five were chosen. slidekit is an agent-first slide builder whose distinctive claim is that layout and aesthetic quality are *provable from source*: a deterministic engine resolves element geometry, and a scorer computes metrics directly over that geometry (element bounding boxes) with no rendering and no vision model. Against the repository's bibliography of 23 verified references and its research trace (which maps every scorer decision to a citation), the five papers below are the ones most tightly coupled to those goals and to the actual code — the primary sources behind verifiable layout rewards, closed-form alignment and overlap metrics, the fourteen geometric aesthetic measures, the non-linear aggregation shipped in the scorer, and the energy-based layout model with learned weights.

Note on scope. Each summary describes the paper on its own terms — its problem, method, and findings — not its relationship to slidekit.

PAPER 1 OF 5

AeSlides: Incentivizing Aesthetic Layout in LLM-Based Slide Generation via Verifiable Rewards

Yiming Pan, Chengwei Hu, Xuancheng Huang, Can Huang, Mingming Zhao, Yuean Bi, Xiaohan Zhang, Aohan Zeng, Linmei Hu

2026 · Preprint

arXiv:2604.22840 · code: github.com/ympan0508/aeslides

Trains an LLM slide generator with reinforcement learning against cheap, checkable layout metrics instead of costly visual inspection.

Seven main points

1. Motivating problem: LLM slide generation is a text-in / text-out process, yet the quality that matters to a viewer is visual and spatial — so optimizing the text objective alone leaves layout aesthetics unaddressed.
2. Prior remedies are expensive or indirect: vision-based “reflection” loops that render and critique each draft, or fine-tuning on aesthetic labels that only indirectly touches layout. AeSlides argues for a direct, quantitative reward instead.
3. Method: a GRPO-style reinforcement-learning setup where the reward is computed from verifiable layout metrics — described as accurate, efficient, and low-cost — rather than from a learned reward model or human/vision feedback.
4. Defines four layout-quality signals it can check automatically: aspect-ratio compliance, whitespace distribution, element collision / overlap, and visual imbalance.

5. The metrics are evaluated on the rendered slide (a browser/Playwright render of the generated layout), turning each into a scalar reward term with no separate reward network in the loop.
6. Results: training on 5,000 prompts with GLM-4.7-Flash lifted aspect-ratio compliance from 36% to 85%, cut whitespace problems ~44%, element collisions ~43%, and visual imbalance ~28%.
7. Human evaluation rose from 3.31 to 3.56 (+7.6%), and the verifiable-reward approach beat both model-based reward optimization and reflection baselines — evidence that deterministic, checkable layout metrics are a viable training signal.

PAPER 2 OF 5

Constrained Graphic Layout Generation via Latent Optimization

Kotaro Kikuchi, Edgar Simo-Serra, Mayu Otani, Kota Yamaguchi

2021 · ACM Multimedia 2021 (pp. 88–96)

DOI:10.1145/3474085.3475497 · arXiv:2108.00871 · code: github.com/ktrk115/const_layout

Generates design layouts that satisfy alignment / non-overlap / relational constraints by optimizing in a generative model's latent space.

Seven main points

1. Addresses automatic graphic-design layout — arranging labeled bounding boxes (text, image, button...) on a canvas — in a way that obeys the design constraints a real designer would impose.
2. Core idea (“latent optimization”): keep a pretrained generative layout model fixed and, at test time, search its latent space so the output satisfies constraints — avoiding a separate retraining for every new constraint type.
3. The generator is a Transformer-based model that produces a layout as a sequence of elements, each with a category label and a bounding box.
4. One model covers many tasks: unconstrained generation, generation with some elements fixed, size/position relationships between elements, and “beautification” of a rough layout.
5. Constraints (alignment, overlap avoidance, and user-specified relations among elements) are expressed as differentiable objectives that steer the latent search while keeping outputs realistic.
6. Uses the now-standard closed-form layout metrics computed purely from bounding boxes: Overlap (average pairwise overlapping area between elements) and Alignment (each element's minimal edge/center misalignment to the others), both normalized by element count.
7. Evaluated on public layout datasets (e.g. Rico, PubLayNet, Magazine); improves constraint satisfaction and layout quality over baselines, with code released for reproduction.

PAPER 3 OF 5

Modelling Interface Aesthetics

David Chek Ling Ngo, Lian Seng Teo, John G. Byrne

2003 · Information Sciences 152:25–46

Proposes 14 closed-form measures that quantify the aesthetics of a screen layout directly from the geometry of its objects.

Seven main points

1. Advances a quantitative theory of interface aesthetics: the visual appeal of a screen layout can be measured mathematically from the positions and sizes of its objects, not just judged subjectively.
2. Defines fourteen aesthetic measures, each a closed-form function of the objects' bounding boxes and each scaled to roughly [0, 1] so layouts can be compared numerically.
3. The fourteen measures: balance, equilibrium, symmetry, sequence, cohesion, unity, proportion, simplicity, density, regularity, economy, homogeneity, rhythm, and order-and-complexity.
4. Balance is the distribution of "optical weight" (element area) about the vertical and horizontal axes; equilibrium places the center of mass near the layout center; symmetry scores axial mirroring of elements.
5. Other measures encode distinct design/Gestalt ideas: cohesion (consistency of element aspect ratios), economy (few distinct sizes), regularity (consistent spacing and alignment), and rhythm (regular repetition across the layout).
6. The authors score real interface layouts with the measures and show they distinguish well-organized arrangements from poor ones, giving worked numeric examples.
7. It became a foundational, heavily-cited reference for computational aesthetics: reproducible formulas that let automated-layout and evaluation systems treat beauty as a computed quantity.

PAPER 4 OF 5

Aesthetic Measures for Automated Document Layout

Steven J. Harrington, J. Fernando Naveda, Rhys Price Jones, Paul Roetling, Nishant Thakkar
2004 · ACM Symposium on Document Engineering (DocEng '04)
DOI:10.1145/1030397.1030419

Gives an automated layout system a computable aesthetic objective, combining geometric measures non-linearly so one flaw dominates the score.

Seven main points

1. Goal: equip an automatic document-layout engine with a computable aesthetic objective so it can rank and choose among candidate page layouts without a human in the loop.
2. Defines heuristic, closed-form measures for the attributes that degrade layout quality: alignment, regularity, separation, balance, white-space fraction, white-space free-flow, proportion, uniformity, and page security.
3. Each measure is normalized so 1 is ideal and lower is worse — e.g. alignment from a histogram of element edge positions, white-space fraction from the proportion of empty page area.

4. Central methodological claim: the individual measures must be combined NON-linearly, not by a plain weighted average that lets a good feature mask a bad one.
5. The proposed aggregation $V = [\sum w_i (d + V_i)^{-p}]^{-1/p} - d$ is a weighted power-mean with a small offset d , so a single very bad sub-measure drags the whole score down toward failure.
6. The weights w_i set the relative importance of each rule, and the exponent p tunes how aggressively the worst feature dominates the combined value.
7. Demonstrated on automated page layout, the paper contributes both a concrete measure set and the enduring argument — adopted by many later systems — that aesthetic aggregation should be non-linear.

PAPER 5 OF 5

Learning Layouts for Single-Page Graphic Designs

Peter O'Donovan, Aseem Agarwala, Aaron Hertzmann

2014 · IEEE Transactions on Visualization and Computer Graphics 20(8):1200–1213

DOI:10.1109/TVCG.2014.48 · project: dgp.toronto.edu/~donovan/layout/

An energy-based model of single-page layout whose geometric terms are weighted by parameters learned from a few example designs.

Seven main points

1. Presents an energy-based model for single-page graphic-design layouts built from established graphic-design principles.
2. The energy is a weighted sum of terms that are functions of element bounding boxes — alignment, overlap, symmetry, and white space / uniform spacing — plus terms for predicted importance and reading flow.
3. Contributes supporting analysis algorithms: prediction of an element's perceived importance, automatic alignment detection, and hierarchical (multi-scale) segmentation of a design.
4. Rather than hand-tuning the term weights, it learns them from a small set of example layouts via Nonlinear Inverse Optimization (NIO).
5. New layouts are produced by optimizing (minimizing) the learned energy, yielding arrangements consistent with the design preferences the examples encode.
6. Demonstrated applications: synthesizing layouts in different styles, retargeting a design to a new page size or aspect ratio, and improving or refining an existing layout.
7. An influential demonstration that design judgment can be captured as a learnable, weighted combination of geometric energy terms and then optimized — linking heuristic aesthetic measures to data-driven design.